

Universal Multimodal Performance Audit High-Precision Audit

Comprehensive Mathematical Compliance & Performance Report
Generated: 2026-04-30 15:18:10

Test Environment

Run Date	Thu Apr 30 14:50:45 CEST 2026
JVM	25

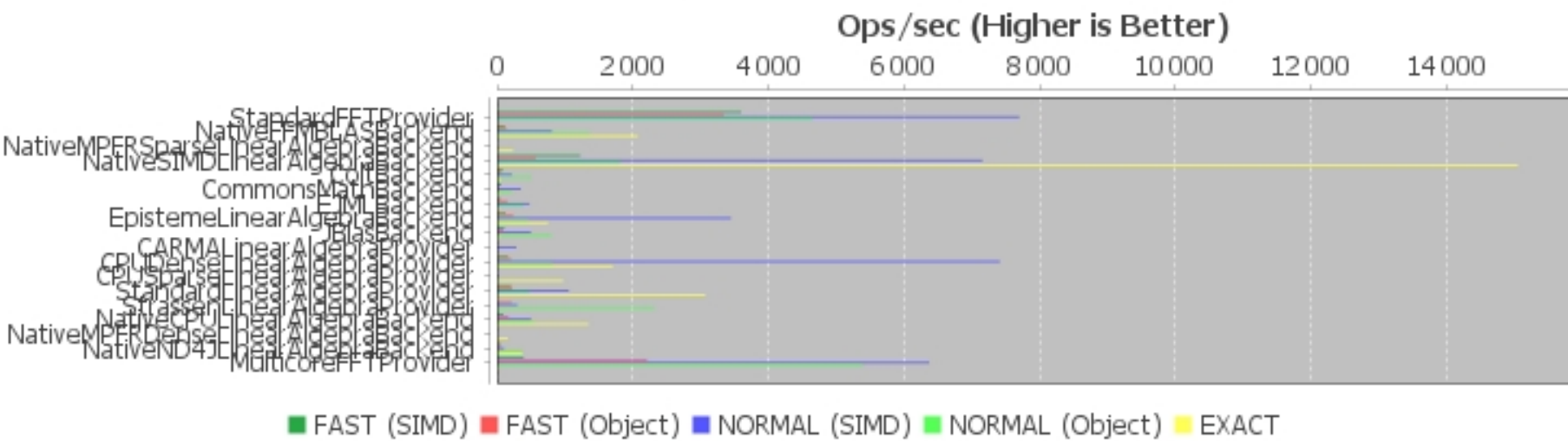
Provider	Domain	Status
MPIDistributedBackend	Distributed Computing	SUCCESS
SparkDistributedBackend	Distributed Computing	SUCCESS
StandardBooleanAlgebraProvider	Boolean Algebra	SUCCESS
StandardFFTProvider	Fast Fourier Transform	SUCCESS
MulticoreGraphAlgorithmProvider	Graph Algorithm	SUCCESS
StandardGraphAlgorithmProvider	Graph Algorithm	SUCCESS
DenseNeuralProvider	Neural Operations	SUCCESS
ONNXBackend	Machine Learning (ONNX)	SUCCESS
ONNXRuntimeBackend	Neural Inference (ONNX)	SUCCESS
MulticoreFEMProvider	Finite Element Method	SUCCESS
RungeKuttaODEProvider	ODE Solver	SUCCESS
MulticoreReduceProvider	reduce	SUCCESS
MulticoreGeneticAlgorithmProvider	Genetic Algorithm	SUCCESS
MulticoreBayesianInferenceProvider	Bayesian Inference	SUCCESS

MulticoreFuzzyProvider	fuzzy logic	SUCCESS
VariableEliminationProvider	Bayesian Inference	SUCCESS
MulticoreMonteCarloProvider	Monte Carlo Simulation	SUCCESS
MulticoreLatticeBoltzmannProvider	Lattice Boltzmann	SUCCESS
MulticoreNavierStokesProvider	Navier-Stokes	SUCCESS
MulticoreSPHFluidProvider	SPH Fluid Dynamics	SUCCESS
MulticoreMolecularDynamicsProvider	Molecular Dynamics	SUCCESS
DistributedNBodyProvider	N-Body Simulation	SUCCESS
MulticoreNBodyProvider	N-Body Simulation	SUCCESS
MulticoreMaxwellProvider	Maxwell Equations	SUCCESS
MulticoreWaveProvider	Wave Equation	SUCCESS
AmazonBraketBackend	Quantum Algorithms	SUCCESS
IBMQBackend	Quantum Algorithms	SUCCESS
PythonQuantumBackend	Quantum Algorithms	SUCCESS
QiskitAerBackend	Quantum Algorithms	SUCCESS
QiskitBackend	Quantum Algorithms	SUCCESS
Quantum4JBackend	Quantum Algorithms	SUCCESS
StrangeBackend	Quantum Algorithms	SUCCESS
MulticoreSCFProvider	Self-Consistent Field	SUCCESS
NativeFFMBLASBackend	Linear Algebra	SUCCESS
NativeMPFRSparseLinearAlgebraBackend	Linear Algebra (Sparse)	SUCCESS
NativeSIMDLinearAlgebraBackend	LinearAlgebra	SUCCESS
NativeND4JTensorBackend	Tensor Operations	SUCCESS
NativeJBulletBackend	Physics/Collision	SUCCESS
NativeQuantumBackend	Quantum Computing	SUCCESS
ColtBackend	Linear Algebra	SUCCESS
CommonsMathBackend	Linear Algebra	SUCCESS
EJMLBackend	Linear Algebra	SUCCESS
EpistemeLinearAlgebraBackend	Linear Algebra (Sparse)	SUCCESS
JBlasBackend	Linear Algebra	SUCCESS
CARMALinearAlgebraProvider	Linear Algebra	SUCCESS

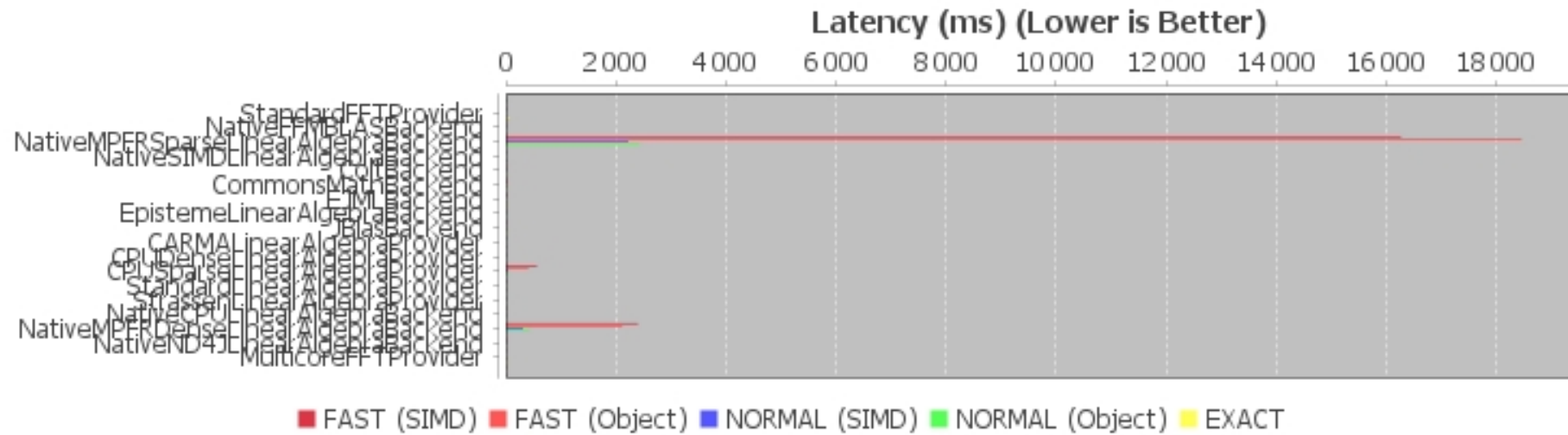
CPU Dense Linear Algebra Provider	Linear Algebra	SUCCESS
CPU Sparse Linear Algebra Provider	Linear Algebra (Sparse)	SUCCESS
Distributed Linear Algebra Provider	Linear Algebra (Sparse)	SUCCESS
Standard Linear Algebra Provider	Linear Algebra	SUCCESS
Strassen Linear Algebra Provider	Linear Algebra	SUCCESS
Native CPU Linear Algebra Backend	Linear Algebra	SUCCESS
Native FFMBLAS Backend	Linear Algebra	SUCCESS
Native MPFR Dense Linear Algebra Backend	Linear Algebra	SUCCESS
Native MPFR Sparse Linear Algebra Backend	Linear Algebra (Sparse)	SUCCESS
Native ND4J Linear Algebra Backend	linear algebra	SUCCESS
Native SIMD Linear Algebra Backend	Linear Algebra	SUCCESS
GRPCL Linear Algebra Backend	Linear Algebra (Sparse)	SUCCESS
Multicore FFT Provider	Fast Fourier Transform	SUCCESS
Standard FFT Provider	Fast Fourier Transform	SUCCESS

Operation Audit: FAST (SIMD)

FAST (SIMD) | Mode Comparison

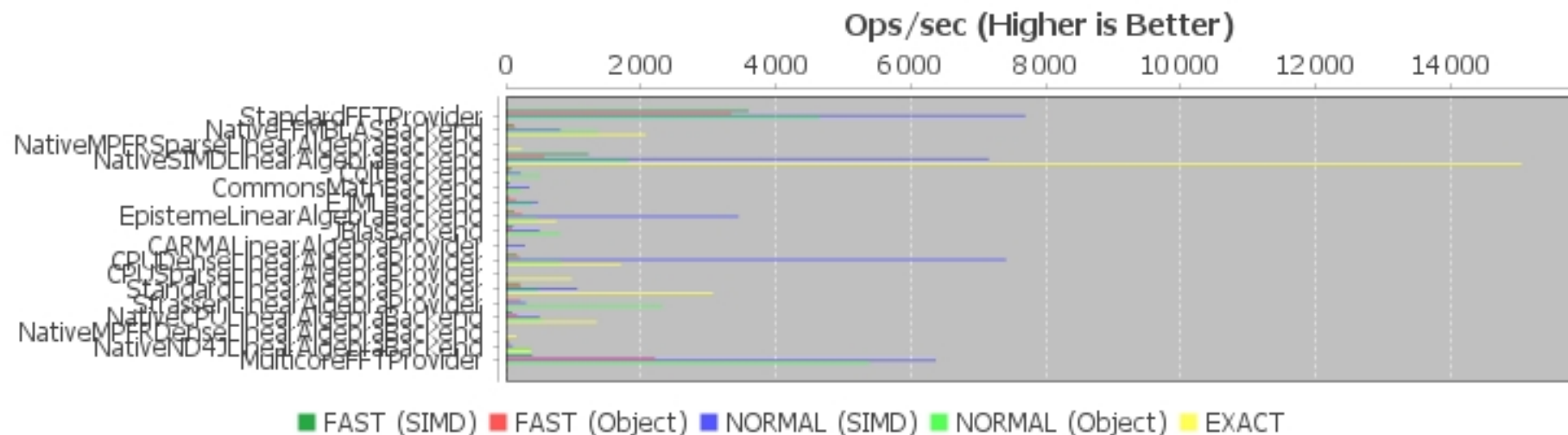


FAST (SIMD) | Mode Comparison

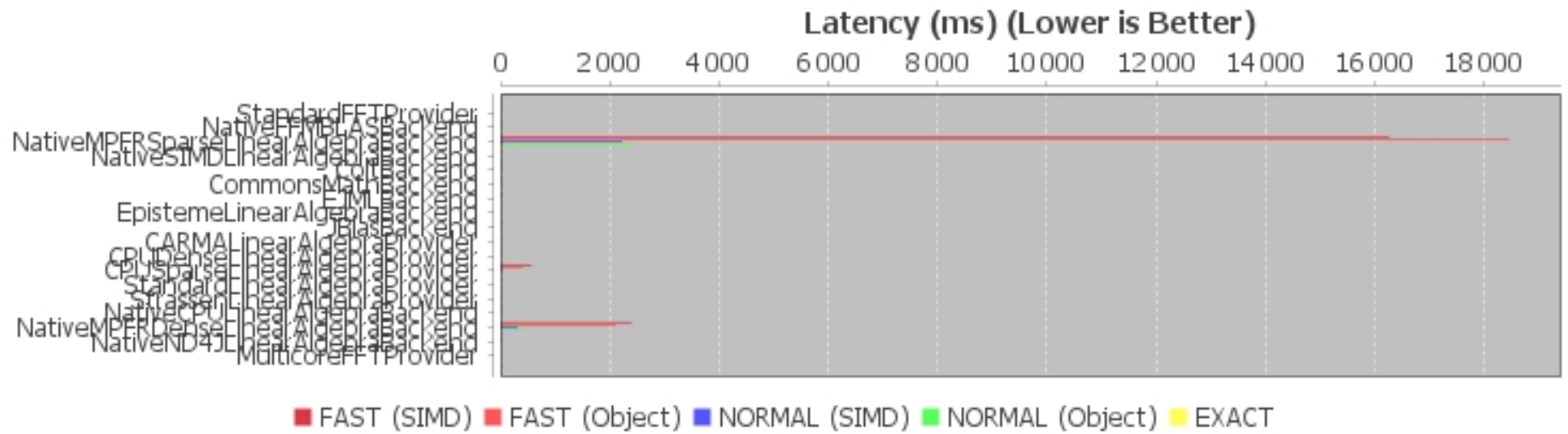


Operation Audit: FAST (Object)

FAST (Object) | Mode Comparison

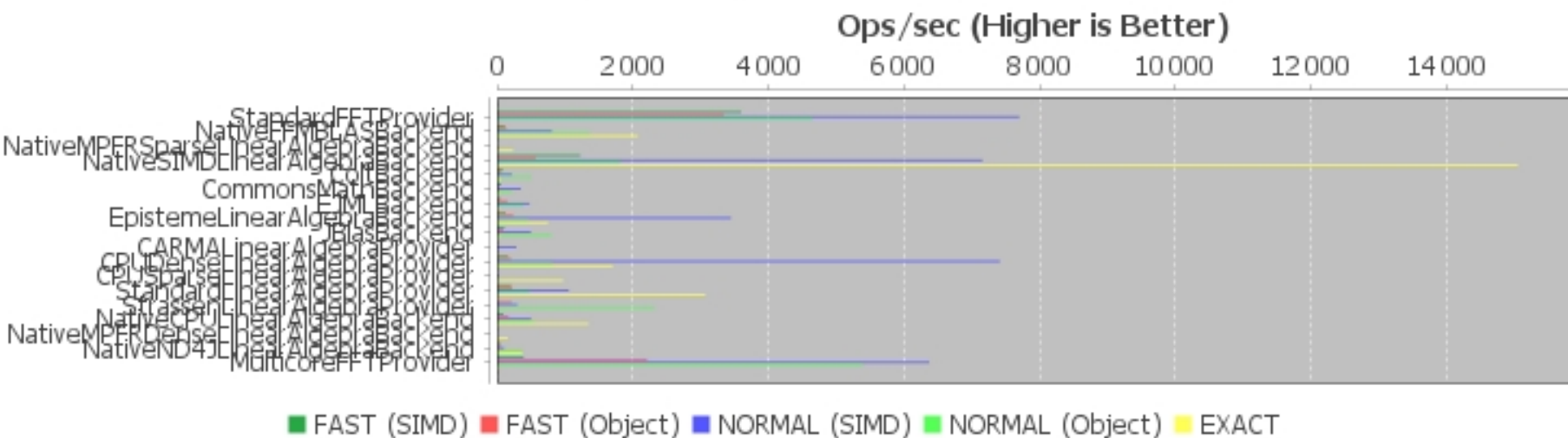


FAST (Object) | Mode Comparison

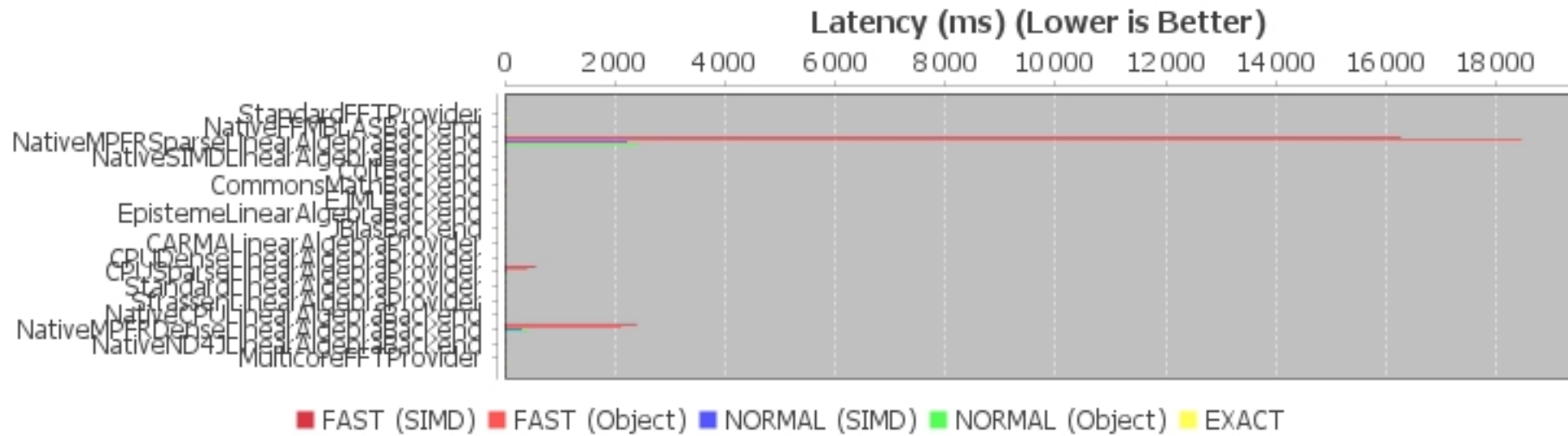


Operation Audit: NORMAL (SIMD)

NORMAL (SIMD) | Mode Comparison

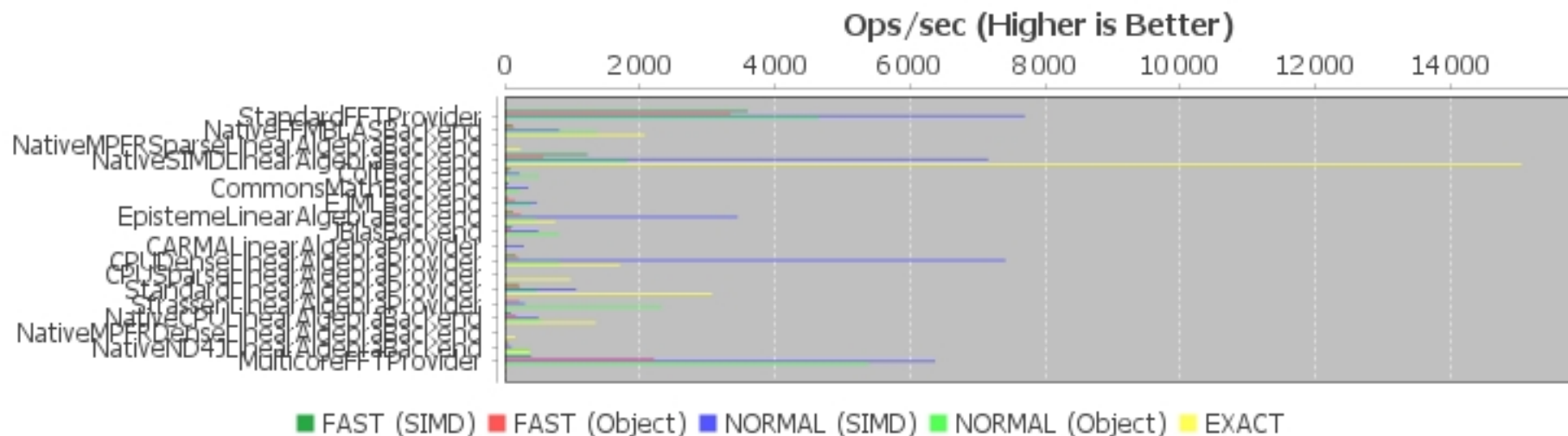


NORMAL (SIMD) | Mode Comparison

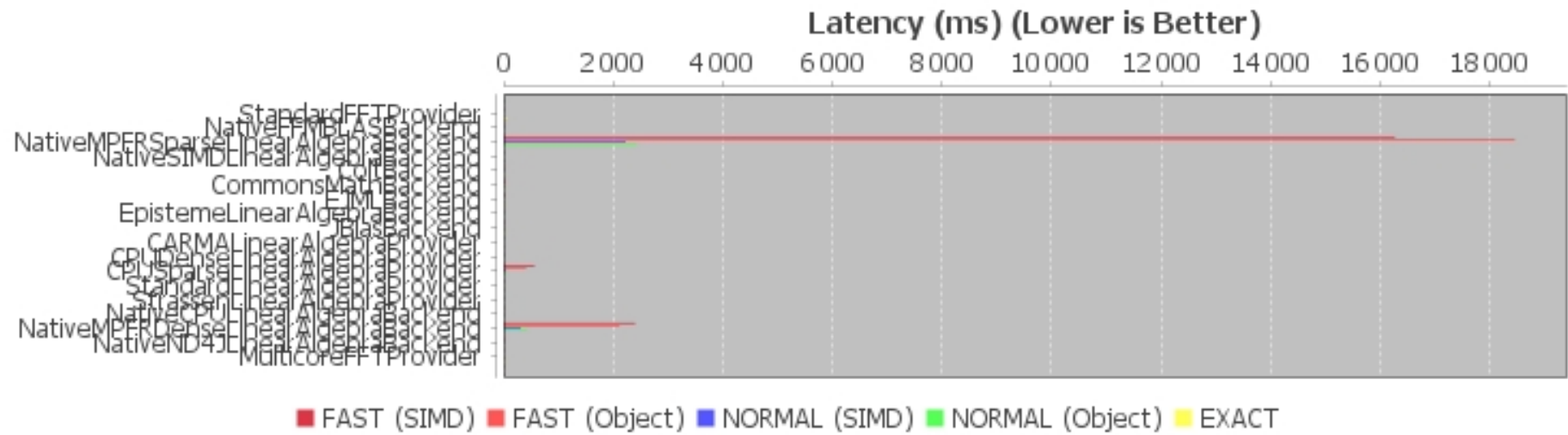


Operation Audit: NORMAL (Object)

NORMAL (Object) | Mode Comparison

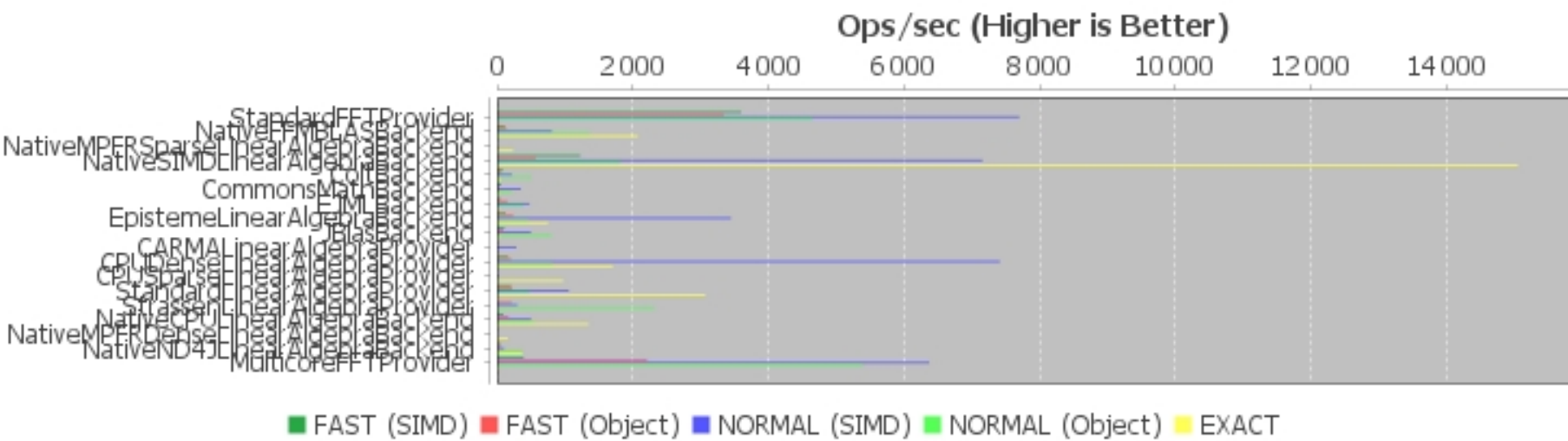


NORMAL (Object) | Mode Comparison



Operation Audit: EXACT

EXACT | Mode Comparison



EXACT | Mode Comparison

